
From Noise to Diversity: Random Embedding Injection in LLM Reasoning

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Abstract

Recent *soft prompt* methods, which prepend or insert trainable continuous embeddings into LLM inputs as virtual tokens, optimize these embeddings to enhance mathematical reasoning. However, the mechanism by which embedding injection itself affects model behavior, independent of optimization, remains unexplored. We investigate Random Soft Prompts (RSPs), continuous vectors sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table and injected without any training. Theoretically, we show that the RSP contribution at each attention layer is exactly captured by a scalar random-token attention mass whose fixed-gap upper envelope shrinks as autoregressive cache growth accumulates real tokens, and that under a local affine analysis of the perturbed logits RSPs can assign positive probability to top- K entry regions that temperature sampling cannot create by rank-preserving scaling. Empirically, RSPs increase early-stage token entropy while generation stabilizes thereafter, and combining RSPs with temperature sampling yields higher trajectory diversity as measured by Pass@ N . We further discuss the implications for GRPO training, where trajectory diversity is critical for learning signal quality.

1 Introduction

As large language models (LLMs) scale to billions of parameters, full fine-tuning becomes increasingly expensive. This has motivated parameter-efficient methods that adapt frozen models by introducing a small number of trainable parameters [Hu et al., 2022, Houlsby et al., 2019]. Among these, *soft prompts* prepend or insert learnable continuous embeddings into the input sequence, steering model behavior through the attention mechanism [Li and Liang, 2021, Lester et al., 2021]. This paradigm has been actively adopted for enhancing mathematical reasoning in LLMs [Kang et al., 2026, Xu et al., 2025, Ye et al., 2026, Hao et al., 2025, Zhang et al., 2026]. Despite their diverse designs, these methods share a common structure: they inject continuous embeddings that lie outside the pretrained vocabulary and optimize them to improve downstream performance.

However, we find that optimization is not a necessary condition for these effects. We observe that injecting randomly initialized, untrained continuous embeddings can also shift the output distribution. For instance, applying a chat template to a base model not fine-tuned with such format causes a prompt-format mismatch that degrades reasoning capability, but appending random embeddings to such mismatched inputs mitigates this degradation. If simple noise were merely perturbing the model, accuracy should worsen instead. This suggests that injecting out-of-vocabulary continuous embeddings itself, independent of any learned content, plays a non-trivial role in shaping model behavior. Existing studies primarily evaluate *soft prompt* methods through end-task accuracy, and the mechanisms by which embedding injection influences the output distribution remain insufficiently explored.

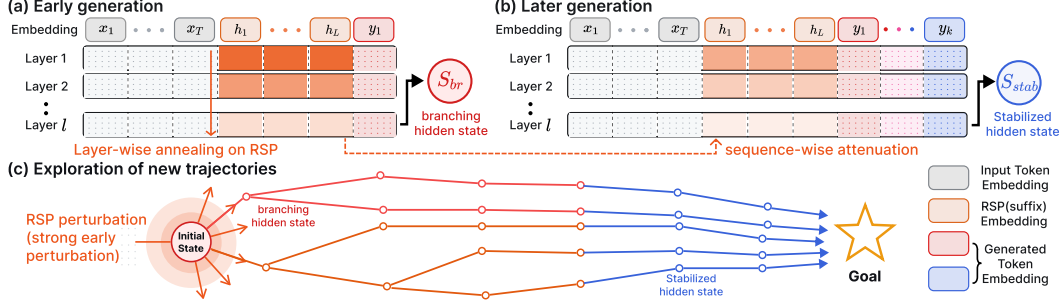


Figure 1: Conceptual overview of RSP-induced trajectory diversity (*suffix injection*). (a) Early decoding: RSP induces a branching hidden state (red) with a diverse output distribution. (b) Later decoding: cache growth attenuates the RSP attention mass, yielding a stabilized hidden state (blue). (c) As generation proceeds, the multiple trajectories opened by early RSP perturbation gradually stabilize into goal-directed completions.

In this work, we investigate the role of Random Soft Prompts (RSPs), continuous vectors sampled from an isotropic Gaussian fitted to the entrywise mean and variance of the pretrained embedding table and injected without any training. The role of randomness is not to imitate the embedding distribution; it is to ensure that repeated rollouts receive *independent* perturbations, which is what exploration requires. Theoretically, we identify an attention-level mechanism in which the RSP contribution is exactly captured by a scalar random-token attention mass whose fixed-gap upper envelope shrinks with autoregressive cache growth, and we show that under a local affine analysis of the perturbed logits RSPs can assign positive probability to top- K entry regions that temperature sampling cannot create by rank-preserving scaling. Empirically, RSPs increase early-stage token entropy while the output stabilizes in later generation stages, and combining RSPs with temperature sampling yields more diverse reasoning trajectories as measured by Pass@ N scaling. We further discuss the implications of these findings for Group Relative Policy Optimization (GRPO) [Shao et al., 2024], where trajectory diversity is essential for effective learning signals.

Our contributions are as follows:

- We give a transformer-level mechanism in which the RSP contribution is controlled by a scalar attention mass with a cache-growth upper envelope, and a distributional analysis under a local affine surrogate showing that *independent isotropic resampling*—not a fixed perturbation—can reach top- K entry regions in a way temperature sampling cannot.
- We empirically analyze how RSPs affect LLM generation, showing that RSPs increase early token entropy and produce more diverse reasoning trajectories when combined with temperature sampling, and that the accumulation gain disappears when one RSP is shared across samples.
- We discuss the implications of RSP-induced diversity for GRPO training and argue that RSP, being rank-changing while temperature is rank-preserving, provides exploration gains complementary to existing temperature-based approaches.

2 Background

2.1 Soft Prompt Methods for LLM Reasoning

Soft prompts emerged as a parameter-efficient alternative to full fine-tuning. Prefix-Tuning [Li and Liang, 2021] introduced the paradigm by prepending trainable continuous vectors to every transformer layer, Prompt Tuning [Lester et al., 2021] simplified this to input-layer-only tuning and showed that it matches full fine-tuning at sufficient scale, and P-tuning [Liu et al., 2024] extended the approach with an LSTM-based prompt encoder to model inter-position dependencies.

More recently, *soft prompts* have been actively applied to LLM reasoning. TTSV [Kang et al., 2026] optimizes prefix embeddings at test time via entropy minimization, steering the model toward high-certainty states to activate latent reasoning capabilities. SoftCoT [Xu et al., 2025] replaces

explicit chain-of-thought tokens with soft tokens generated by an assistant model, leveraging the stronger representational capacity of continuous representations over discrete language. LTPO [Ye et al., 2026] optimizes latent thought embeddings per test instance through policy gradient with a confidence-based intrinsic reward. COCONUT [Hao et al., 2025] feeds the model’s own hidden states back as input embeddings, enabling reasoning in a continuous latent space. MemGen [Zhang et al., 2026] demonstrates that embedding vectors can serve as experience memory by constructing latent token sequences that enrich the model’s reasoning. Pause tokens [Goyal et al., 2024] insert learnable tokens into the input to provide additional computation steps.

These approaches all inject out-of-vocabulary continuous embeddings optimized for task-specific objectives. Because evaluation centers on end-task accuracy, the contribution of the injection itself and that of the optimization are not separated, and the mechanisms by which such out-of-vocabulary continuous embeddings shape model behavior remain a separate and underexplored question.

2.2 Noise Injection in Neural Networks

Neural networks have incorporated noise at various stages. During training, dropout [Srivastava et al., 2014] randomly deactivates hidden units to prevent overfitting, and NEFTune [Jain et al., 2024] adds noise to token embeddings during instruction fine-tuning, reporting improvements in instruction following. These methods use noise as a regularizer and do not apply perturbations at inference time.

At inference time, perturbation-based approaches primarily target robustness and safety. Randomized smoothing [Cohen et al., 2019] injects Gaussian noise into inputs to obtain certified robustness guarantees. In the LLM setting, SmoothLLM [Robey et al., 2025] perturbs input prompts at the character level to defend against adversarial jailbreaking. These methods require multiple noisy forward passes with output aggregation, and target prediction stability rather than output diversity.

In reinforcement learning, NoisyNet [Fortunato et al., 2018] injects learnable noise into network weights to improve exploration, demonstrating that the location and structure of noise injection affect exploration quality. This motivation is closest to RSPs, which similarly target diversity, but differ in injection target (input embeddings vs. network weights) and require no learned noise parameters.

RSPs differ from these approaches in the following respects. (1) RSPs operate solely at inference without any training, whereas dropout and NEFTune inject noise during training, modifying the resulting model parameters accordingly. (2) RSPs preserve the original input and append fresh embedding vectors in a single forward pass, whereas robustness methods corrupt the input and aggregate over multiple noisy passes. (3) RSPs use purely random vectors sampled from the pretrained embedding statistics, whereas NoisyNet learns noise parameters through optimization.

3 Random Soft Prompts and Why They Work

This section states the mechanism used in the paper. A *rollout* means one full autoregressive decode under one sampled RSP. Within a rollout, an RSP enters the hidden state only through attention; the same scalar attention mass that enables an early influence is constrained by a cache-growth envelope as real tokens accumulate. Across rollouts, independent resampling can accumulate early branch differences into Pass@N gains under the task-side assumptions stated below. We keep the main statements and intuition here; further derivations are in Appendix D.

3.1 Random Soft Prompt

We start with the object being analyzed. Let $\mathbf{W}_E \in \mathbb{R}^{V \times d}$ denote the pretrained token-embedding matrix, where V is the vocabulary size and d is the hidden dimension. Write its entrywise mean and standard deviation as $\mu_E := \frac{1}{Vd} \sum_{i,k} [\mathbf{W}_E]_{i,k}$ and $\sigma_E := \text{std}(\mathbf{W}_E)$. An RSP of length L is a sequence of continuous vectors $\mathbf{H} = [h_1, \dots, h_L] \in \mathbb{R}^{L \times d}$ drawn independently from

$$h_j \sim \mathcal{N}(\mu_E \mathbf{1}_d, \sigma_E^2 \mathbf{I}_d), \quad j = 1, \dots, L. \quad (1)$$

The centered form $\bar{h}_j := h_j - \mu_E \mathbf{1}_d \sim \mathcal{N}(0, \sigma_E^2 \mathbf{I}_d)$ is a zero-mean isotropic Gaussian. The RSP can be concatenated with the input embeddings $\mathbf{X}$ at one of three positions: *prefix* ($[\mathbf{H}; \mathbf{X}]$), *infix* ($\mathbf{H}$ inserted within $\mathbf{X}$), or *suffix* ($[\mathbf{X}; \mathbf{H}]$). These positions reflect the injection strategies in prior work: TTSV [Kang et al., 2026] prepends as *prefix*, SoftCoT [Xu et al., 2025] and LTPO [Ye et al., 2026]

120 insert as *infix* between user and assistant turns, and MemGen [Zhang et al., 2026] appends as *suffix*.
 121 For repeated-sampling protocols, RSP draws and decoding randomness are independent across trials.
 122 The role of this randomness is not to imitate the embedding distribution but to give repeated rollouts
 123 *independent* perturbations, which is what exploration requires.

124 3.2 Exploration: how RSP enters the hidden state

125 We first ask how a single RSP injection perturbs the hidden state. RSP does not add to the hidden
 126 state directly; it enters only through attention weights, so its instantaneous influence within one head
 127 can collect into a single quantity: the attention mass assigned to RSP tokens.

128 Concretely, consider one attention head in self-attention layer ℓ with $n \geq 1$ unmasked real tokens,
 129 $L \geq 1$ RSP tokens, finite attention logits, attention weights $\alpha_{ij}^{(\ell)}$, and value vectors $\mathbf{v}_j^{(\ell)}, \mathbf{v}_{r_j}^{(\ell)}$ on the
 130 real and RSP sides. Define the RSP attention mass $w_{r,i}^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)}$. Since the real-token mass
 131 is then positive, the head output decomposes *exactly* as

$$\mathbf{o}_i^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)} + \boldsymbol{\eta}_i^{(\ell)}, \quad \tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^n \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)}, \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}. \quad (2)$$

132 *Derivation of Eq. (2).* Write the head output as $\sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}$, then factor
 133 $1 - w_{r,i}^{(\ell)} = \sum_{j=1}^n \alpha_{ij}^{(\ell)} > 0$ out of the real-token sum. No approximation beyond the softmax-
 134 normalization identity is used. $\square$

135 The same scalar $w_{r,i}^{(\ell)}$ controls two things at once. The attenuation ratio $1 - w_{r,i}^{(\ell)}$ on the real-token
 136 signal and the upper bound $\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_j \|\mathbf{v}_{r_j}^{(\ell)}\|$ on the random contribution are both tied to
 137 it. The RSP-induced variation reduces to tracking a single number in $[0, 1]$. Early in decoding the
 138 cache is short and $w_{r,i}^{(\ell)}$ can be nontrivial, so the same identity that bounds the random contribution
 139 also lets one sampled RSP perturb early next-token decisions.

140 3.3 Annealing: KV cache growth dilutes RSP attention

141 The natural next question is how this scalar behaves as decoding proceeds. In autoregressive
 142 generation, the KV cache adds one real-token term at each step while the number of RSP terms stays
 143 fixed. Under a fixed real-vs-random logit gap, this cache growth gives an upper envelope.

144 **Theorem 1** (Attention-mass decay under cache growth). *For a query attending $n \geq 1$ real tokens
 145 and $L \geq 1$ RSP tokens with finite logits, let $\Delta_i^{(\ell)} := \min_{j \in [n]} \mathbf{q}_i^\top \mathbf{k}_j - \max_{j' \in [L]} \mathbf{q}_i^\top \mathbf{k}_{r_{j'}}$, denote the
 146 pre-scale logit gap between real and random tokens. In scaled dot-product attention,*

$$w_{r,i}^{(\ell)} \leq \frac{L}{L + n \exp(\Delta_i^{(\ell)} / \sqrt{d})},$$

147 *and the right-hand side is strictly decreasing in n and tends to 0 at fixed L and $\Delta_i^{(\ell)}$.*

148 *Proof.* Let $s_j := \mathbf{q}_i^\top \mathbf{k}_j / \sqrt{d}$, $s_{r_{j'}} := \mathbf{q}_i^\top \mathbf{k}_{r_{j'}} / \sqrt{d}$, $s_{r,\max} := \max_{j'} s_{r_{j'}}$, and $R := \sum_{j'=1}^L \exp(s_{r_{j'}})$.
 149 The gap definition gives $s_j \geq \Delta_i^{(\ell)} / \sqrt{d} + s_{r,\max}$ for every real j , and $\exp(s_{r,\max}) \geq R/L$ holds
 150 because the max dominates the average; combining the two and summing over the n real keys yields
 151 $\sum_{j=1}^n \exp(s_j) \geq (n/L) \exp(\Delta_i^{(\ell)} / \sqrt{d}) R$. Substituting into $w_{r,i}^{(\ell)} = R / (\sum_j \exp(s_j) + R)$ gives

$$w_{r,i}^{(\ell)} \leq \frac{R}{(n/L) \exp(\Delta_i^{(\ell)} / \sqrt{d}) R + R} = \frac{L}{L + n \exp(\Delta_i^{(\ell)} / \sqrt{d})}.$$

152 The derivative of the right-hand side in n is $-L \exp(\Delta_i^{(\ell)} / \sqrt{d}) / (L + n \exp(\Delta_i^{(\ell)} / \sqrt{d}))^2 < 0$, so the
 153 bound is strictly decreasing and tends to 0 as $n \rightarrow \infty$. $\square$

154 This single inequality carries two messages. The RSP length L is the design parameter we choose;
 155 it sits in the numerator and sets the *maximum* influence of the perturbation. The real-token count n
 156 grows automatically every step and sits in the denominator, so the *envelope* of the influence shrinks
 157 under the same L . What the theorem guarantees is the sequence-direction attenuation of the upper
 158 envelope at fixed (layer, query position, logit gap), not a layer-direction monotone decay; queries,
 159 keys, and hidden states co-evolve, so $\Delta_i^{(\ell)}$ also moves and the realized $w_{r,i}^{(\ell)}$ need not decrease at
 160 every step. The accurate reading is that “the envelope compatible with a fixed gap shrinks as n grows.”
 161 Because Eq. (2) bounds the random contribution by the same scalar, the perturbation magnitude
 162 inherits the same envelope. This gives an implicit explore-then-exploit interpretation of decoding
 163 without asserting stepwise monotonic decay. Figure 2 verifies the sequence-direction prediction
 164 empirically.

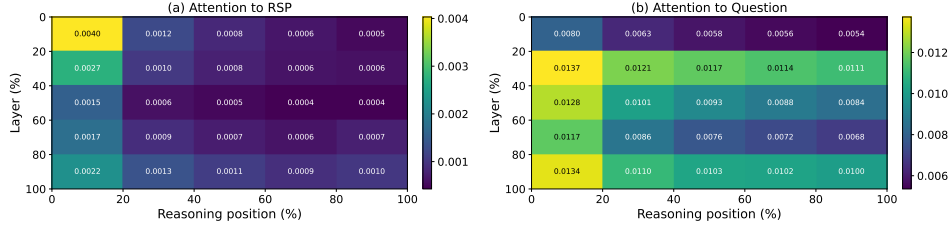


Figure 2: Per-token attention mass on Qwen2.5-Math-7B (suffix, 500 MATH-500 problems, each with an independently sampled RSP). (a) RSP-token attention decreases along the reasoning-position axis, consistent with the cache-growth bound of Theorem 1. The additional decrease along the layer axis is a separate empirical phenomenon suggesting that deeper layers sharpen the real-vs-random logit gap; it does not follow from Theorem 1 alone. (b) Question-token attention remains comparatively uniform. The reported quantity is the per-token mass of Appendix E, equal to $w_{r,i}^{(\ell)}/L$ averaged over heads and samples.

165 3.4 Performance gain: why independent reruns help

166 Theorem 1 explains what one rollout can do. The remaining question is performance: why can *fresh*
 167 RSP draws across reruns help more than repeating a single fixed perturbation? The answer needs one
 168 distributional fact about the prompt law and one separate task-side fact about productive branches.

169 The first fact is directional. The attention identity above does not require isotropy, but branch opening
 170 benefits from a perturbation law that does not under-serve whichever local direction matters for the
 171 current prompt. We state this fact for one centered RSP vector; the local surrogate below isolates one
 172 such perturbation from the length- L prompt. A minimax statement formalizes the point.

173 **Theorem 2** (Direction-agnostic minimax coverage). *Let D be any zero-mean distribution on $\mathbb{R}^d$ with*
 174 *covariance Σ_D and trace budget $\text{tr}(\Sigma_D) \leq \rho^2$. Then $\min_{\|b\|=1} \text{Var}_{h \sim D}(b^\top h) = \lambda_{\min}(\Sigma_D) \leq \rho^2/d$,*
 175 *with equality if and only if $\Sigma_D = (\rho^2/d)\mathbf{I}_d$.*

176 Equality holds for any distribution with isotropic covariance—Gaussianity is not the unique answer.
 177 The Gaussian RSP design *borrow*s the scale from the embedding distribution but discards its direc-
 178 tional geometry on purpose. The isotropic Gaussian additionally has positive density on all of $\mathbb{R}^d$
 179 (full support), so it places no prior weight on any direction (Appendix C); learned fixed prompts
 180 ($\Sigma_D = 0$), PCA-aligned noise, vocabulary-token sampling, and embedding-covariance matching are
 181 all anisotropic and so under-explore at least one direction.

182 We now formalize branch opening on the surrogate. Linearizing the logits around $\bar{h} = 0$, define
 183 the affine surrogate $z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}$ with $a_i := \nabla_{\bar{h}} z_i(0)$. For a baseline-excluded token i , the
 184 top- K entry region is $\mathcal{G}_{i,K} := \{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K - 1\}$, matching the
 185 actual top- K definition.

186 **Proposition 1** (Conditional top- K entry). *If $\mathcal{G}_{i,K}$ contains a nonempty open set, the isotropic*
 187 *Gaussian’s full support assigns it positive probability, so $\mu_i := \Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) > 0$.*

188 This proposition gives a conditional reachability statement: once such a local top- K region exists
 189 in the surrogate, the RSP law assigns it positive probability. The existence of that region remains

Table 1: Accuracy (%) across model configurations and benchmarks. RSP reports the best result across injection positions (Prefix, Infix, Suffix). Full per-position breakdown is in Appendix A. Values in parentheses indicate the difference from the baseline.

Model	MATH-500		GSM8K		AIME24	
	Baseline	RSP	Baseline	RSP	Baseline	RSP
<i>Instruct Models</i>						
LLaMA-3.1-8B-Inst	52.20	49.40 (−2.8)	85.75	86.73 (+1.0)	6.67	10.00 (+3.3)
Qwen2.5-Math-7B-Inst	83.20	84.20 (+1.0)	95.45	95.68 (+0.2)	13.33	16.67 (+3.3)
Qwen2.5-Math-1.5B-Inst	73.00	75.20 (+2.2)	85.29	85.75 (+0.5)	10.00	20.00 (+10.0)
<i>Base Models (Raw Text)</i>						
Qwen2.5-Math-7B	72.20	71.60 (−0.6)	84.15	84.31 (+0.2)	6.67	16.67 (+10.0)
Qwen2.5-Math-1.5B	61.40	64.40 (+3.0)	77.86	74.30 (−3.6)	16.67	10.00 (−6.7)
<i>Base Models + ChatML (Format Mismatch)</i>						
Qwen2.5-Math-7B	52.20	70.40 (+18.2)	58.30	75.97 (+17.7)	23.33	23.33 (+0.0)
Qwen2.5-Math-1.5B	34.40	63.40 (+29.0)	37.45	67.02 (+29.6)	13.33	20.00 (+6.7)

a task-side condition, not a consequence of the mechanism alone. To lift a single-draw opening probability to task accuracy we add two task-side assumptions: (M1) each independent RSP run reaches a correct trajectory with marginal probability at least $p_{\min}(x) > 0$, and (M2) the N runs are mutually independent.

Proposition 2 (Multi-path Pass@ N). *Under (M1)–(M2), with N independent RSP runs, $\Pr(\text{Pass}@N \text{ on } x) \geq 1 - (1 - p_{\min}(x))^N \geq 1 - \exp(-N p_{\min}(x))$.*

This division of labor is the main interpretive point. The mechanism can *admit* baseline-excluded candidates into the local candidate set when the corresponding entry region exists; whether an admitted branch is *productive* is task-dependent, which is why $p_{\min}(x) > 0$ is a separate assumption. Sharing one RSP across all samples removes this accumulation, so *independence* is essential. Section 4 tests exactly this signature.

4 Empirical Validation

We now examine how the explanation in §3 appears in actual LLM generation through three scenes. First, we ask whether RSP preserves baseline accuracy without breaking it, and recovers it in some settings (§4.2). Second, we check whether early diversification followed by late stabilization—the empirical signature suggested by the attention decomposition and cache-growth envelope—is observed (§4.3). Finally, we contrast Pass@ N between sharing one RSP across samples versus drawing a fresh RSP per sample, directly probing whether *independence* is the key (§4.4).

4.1 Experimental Setup

We evaluate on three mathematical reasoning benchmarks, MATH-500 [Lightman et al., 2024], GSM8K [Cobbe et al., 2021], and AIME24, using LLaMA-3.1-8B-Instruct [Grattafiori et al., 2024] and the Qwen2.5-Math model family [Yang et al., 2024] across instruct models, base models, and base models with mismatched chat templates—seven configurations in total. All experiments use greedy decoding to isolate the effect of RSPs from sampling stochasticity, and the answer-extraction pipeline uses fallbacks to score only the final answer. RSPs follow the definition of §3.1 and are concatenated at one of three positions (prefix, infix, suffix) with a freshly drawn $\mathbf{H}$ for each batch. Full experimental details and per-position results are in Appendix A.

4.2 How Does Untrained RSP Compare to Baselines & Optimized Soft Prompts?

We first examine how injecting random embeddings affects the model’s reasoning performance. Table 1 reports the accuracy for each model configuration, using the best RSP result across injection positions (Prefix, Infix, Suffix) and $L \in \{10, 15, 20\}$ for each benchmark.

Table 2: Comparison with existing soft prompt methods (TTSV, SoftCoT, LTPO) under unified evaluation. Baseline and RSP values are from Table 1. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Benchmark	Baseline	RSP	TTSV	SoftCoT	LTPO
Qwen2.5-Math-7B-Instruct	MATH-500	83.20	84.20	<u>83.80</u>	82.80	83.00
	GSM8K	<u>95.45</u>	95.68	95.30	95.00	94.84
	AIME24	13.33	<u>16.67</u>	23.30	<u>16.67</u>	13.33
Qwen2.5-Math-1.5B-Instruct	MATH-500	73.00	<u>75.20</u>	<u>75.20</u>	76.00	72.40
	GSM8K	85.29	85.75	<u>85.70</u>	84.46	84.53
	AIME24	<u>10.00</u>	20.00	6.70	6.67	<u>10.00</u>

For instruct models and base models with their native prompt formats, RSP injection maintains or slightly shifts baseline performance within a few percentage points. Most strikingly, under prompt-format mismatch — where base models receive ChatML templates they were not trained on — RSP yields pronounced improvements of up to +29 pp. These gains are difficult to attribute to simple perturbation effects, as noise that merely raises model uncertainty would further degrade performance.

We compare RSPs with prior soft prompt methods (TTSV, SoftCoT, LTPO) that optimize their embeddings for distinct objectives, under a unified evaluation protocol reported in Table 2. All methods are re-evaluated with the same unified extraction pipeline (Appendix A.2), enabling a format-agnostic comparison despite each method’s distinct prompt format and grading scheme. The results place RSPs on par with the optimized methods without requiring any training.

4.3 How Does RSP Affect the Output Distribution?

We analyze how RSP injection changes the model’s output distribution on Qwen2.5-Math-1.5B-Instruct with MATH-500, measuring per-token entropy, top-1 probability, and varentropy across the generation process. Figure 3 compares these metrics for the first 5% of generation steps across RSP variants and prior soft prompt methods (LTPO, SoftCoT, TTSV). Full statistics are in Appendix A.4.

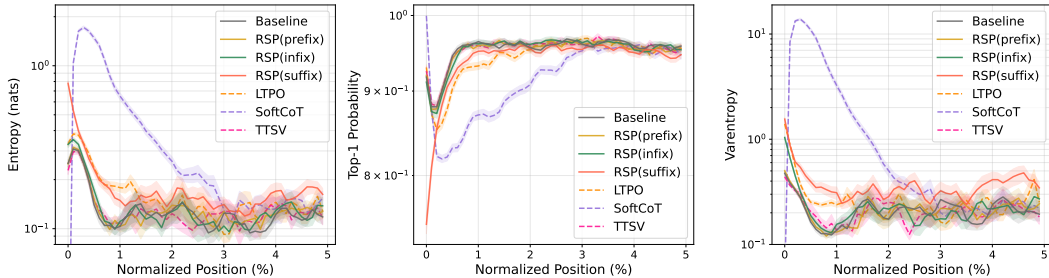


Figure 3: Mean entropy, top-1 probability, and varentropy during the first 5% of generation steps (Qwen2.5-Math-1.5B-Instruct, MATH-500). Shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: prior soft prompt methods (LTPO, SoftCoT, TTSV).

These results show that RSP modifies the early-stage output distribution across all injection positions: entropy increases (the token distribution flattens), top-1 probability decreases (commitment to a dominant token weakens), and varentropy increases (indicating multimodal distributions [Ahmed et al., 2026]). High-entropy positions are known to act as critical forking points that determine reasoning trajectories [Wang et al., 2025], so this shift moves the model toward a state where multiple reasoning paths can coexist. The change is localized to the initial reasoning stage. In the middle and final portions of generation, all three metrics converge to baseline levels.

Although existing soft prompt methods are optimized for different objectives, they share a common pattern of increasing early-stage entropy. Notably, TTSV, which optimizes soft prompts using a reward that reduces the mean entropy of reasoning trajectories, still exhibits early entropy comparable to the baseline. This suggests that elevating early-stage entropy is an inherent effect of injecting non-pretrained continuous embeddings, independent of the optimization objective.

4.4 Does RSP Induce Trajectory Diversity?

Section 4.3 empirically established that RSP reshapes the early-stage output distribution. We now examine whether this shift translates into reasoning diversity at three complementary levels of analysis: token rankings, semantic content of trajectories, and task-level outcomes. The three analyses converge on a directional picture: RSP’s first-token perturbation propagates into semantically more diverse reasoning trajectories, which in turn yields greater output diversity at the task level. All three analyses share the setting: Qwen2.5-Math-1.5B-Instruct, MATH-500, suffix injection, and $L = 10$.

Token-level: distribution beyond temperature.

We quantify how much RSP shifts the first-token distribution, using temperature sampling as a baseline. Under autoregressive decoding, any trajectory-level divergence between RSP and the baseline must originate from the first generation step, a critical forking point for reasoning trajectories [Wang et al., 2025]. With the baseline at $\tau=1.0$ as reference, we compare baselines at $\tau=2.0$ and $\tau=3.0$ (16 samples per problem) against RSP at $\tau=1.0$ (averaged over 16 seeds) on four metrics: Spearman rank correlation ρ measuring whether the relative ranking of tokens is preserved, probability mass outside the baseline’s top-10 measuring support expansion to low-probability tokens, Jensen–Shannon divergence as the overall distributional distance, and Pass@1 accuracy (formal definitions in Appendix B).

Table 3: First-token distribution metrics measured against the baseline at $\tau=1.0$, comparing baselines at higher temperatures ($\tau=2.0, 3.0$) with RSP at $\tau=1.0$. RSP metrics are averaged over 16 random seeds; Acc reports mean $\pm$ std across 16 samples per problem.

Metric	$\tau=2.0$	$\tau=3.0$	RSP
Spearman ρ	1.000	1.000	0.709
Mass outside top-10	5.18%	29.11%	21.86%
JS divergence	0.060	0.218	0.131
Acc (%)	20.77 ± 1.47	1.42 ± 0.35	73.30 ± 1.07

Table 3 reports the contrast. RSP’s distributional shift falls between $\tau=2.0$ and $\tau=3.0$ in magnitude (mass outside top-10 and JS divergence), yet its mechanism and cost differ sharply. Temperature scaling is a monotone transform that preserves token ranking, so ρ relative to the baseline is exactly 1 at any τ . RSP partially preserves the ranking but induces reranking, yielding $\rho = 0.709$. The Pass@1 row exposes the cost: approaching RSP’s magnitude requires raising τ toward 3.0, but accuracy collapses to 1.42%, while RSP preserves accuracy at 73.30%. Rather than uniformly flattening the distribution as temperature scaling does, RSP produces a fundamentally different kind of perturbation.

Trajectory-level: semantic diversity.

Token-level reranking raises a second question: do first-token perturbations produce semantically diverse reasoning trajectories, or do independently perturbed trajectories converge to semantically similar ones? We randomly sample 64 problems from MATH-500 and generate 300 independent trajectories per problem under Baseline and RSP at temperature $\tau = 1.0$. Each trajectory is encoded into a dense semantic vector by BGE-M3 [Chen et al., 2024], a sentence-level multilingual embedding model, and we compute three complementary metrics over these embeddings: *pairwise cosine distance* between trajectories captures the overall semantic spread of the trajectory set; *inter-cluster distance* between centroids identified by DBSCAN [Ester et al., 1996], a density-based clustering algorithm that groups nearby trajectories without prespecifying the number of clusters, captures the separation between distinct trajectory groups; and *intra-cluster distance* captures semantic coherence within each group.

Table 4: Semantic diversity metrics (64 problems, 300 trajectories per condition).

Metric	Baseline	RSP
Pairwise cos. dist.	0.0612	0.0879
Inter-cluster dist.	0.0183	0.0982
Intra-cluster dist.	0.0526	0.0528

Table 4 reveals three concurrent patterns. Pairwise cosine distance rises by $1.4\times$, indicating that the trajectory set as a whole becomes more semantically diverse. Inter-cluster distance jumps by $5.4\times$, showing that the trajectories split into substantially more separated groups. Intra-cluster distance is essentially unchanged, indicating that semantic coherence within each group remains comparable to the baseline. RSP also yields larger pairwise distance than baseline on 56 of 64 problems (87.5%).

Outcome-level: Pass@N scaling.

Finally, we evaluate whether the token- and trajectory-level diversity translates into task-level outcome diversity through Pass@N [Chen et al., 2021], the probability that at least one of N sampled solutions is correct. We compare four conditions with $N = 16$ samples per problem on MATH-500 and AIME24: (i) *Baseline*, $\tau = 1.0$: temperature sampling only; (ii) *RSP (fixed seed)*, $\tau = 1.0$: a single RSP shared across all samples; (iii) *RSP (indep.*

seed), greedy: a different RSP per sample without temperature; and (iv) RSP (indep. seed), $\tau = 1.0$: a different RSP per sample combined with temperature sampling.

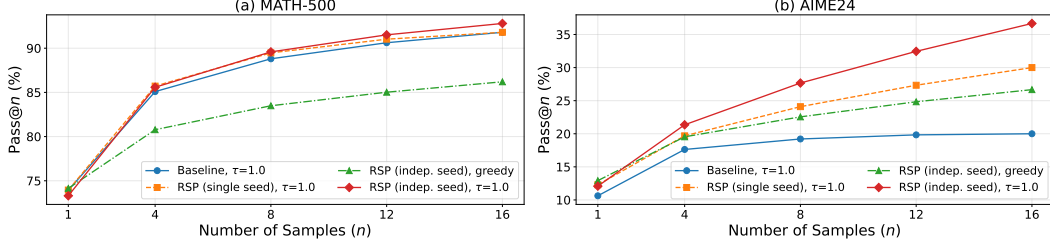


Figure 4: Pass@ N scaling on (a) MATH-500 and (b) AIME24 with Qwen2.5-Math-1.5B-Instruct, $N = 16$ samples per problem. Baseline: temperature sampling only. Fixed seed: single RSP shared across samples combined with temperature. Indep. seed: a different RSP per sample, with or without temperature.

Condition (iv) achieves the highest Pass@ N on both benchmarks, reaching 92.80% on MATH-500 and 36.67% on AIME24 at $N = 16$. Condition (ii) shows no improvement over the baseline, suggesting that the diversity gain likely depends on varying the random embeddings across samples rather than a single fixed perturbation. Pass@1 remains comparable to the baseline on MATH-500, and on the more challenging AIME24, (iv) achieves higher Pass@1 than the baseline. Together these observations suggest that combining independent RSPs with temperature sampling may yield greater trajectory diversity while preserving single-sample accuracy.

5 Application: DAPO Training with RSP

The empirical analysis so far evaluates RSP at inference time only. Section 3.4 predicts that independent RSP draws should benefit any sampling-based optimization that depends on rollout diversity. We test this prediction by integrating RSP into a GRPO-based RL training pipeline. Implementation details and full per-step results are deferred to Appendix F.

DAPO. DAPO [Yu et al., 2025] extends GRPO [Shao et al., 2024] with loss-term modifications that preserve rollout diversity during long chain-of-thought RL training. The diversity is shaped at the loss level by reshaping how rollouts are weighted and clipped. RSP, in contrast, induces diversity at the input level by perturbing the rollout distribution before any loss is computed (Section 4.4). We use DAPO as a testbed to ask whether these two diversity sources compose to yield further performance gains. Let o_i be the i -th rollout response, $r_{i,t}(\theta) = \pi_\theta(o_{i,t} | q, [\mathbf{H}_g,]o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} | q, [\mathbf{H}_g,]o_{i,<t})$ the importance ratio, $\hat{A}_{i,t}$ the group-relative advantage, and $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}})$ the asymmetric clip range. The bracketed $\mathbf{H}_g$ is the RSP appended to the prompt. The DAPO + RSP objective is

$$\mathcal{J}_{\text{DAPO+RSP}}(\theta) = \mathbb{E} \left[\frac{1}{\sum_i |o_i|} \sum_{i,t} \min \left(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \right]. \quad (3)$$

Setup. We train Qwen2.5-Math-7B on a MATH Level-3–5 subset (~8.5K prompts) using the SimpleRL-Zoo [Zeng et al., 2025] training pipeline with the objective of Eq. (3). Each step samples $G = 8$ rollouts per prompt over a batch of 1,024 prompts under $\tau=1.0$, for 100 training steps (approximately 12 epochs). We compare DAPO against DAPO + RSP, where RSP suffix injection ($L = 20$ tokens) is applied during rollouts only; evaluation is performed without RSP. We evaluate on seven mathematical-reasoning benchmarks (MATH-500 [Lightman et al., 2024], AIME 2024, AMC 2023, GSM8K [Cobbe et al., 2021], GaoKao 2023-EN, OlympiadBench [He et al., 2024], Minerva Math [Lewkowycz et al., 2022]). The small competition sets (AIME 2024, AMC 2023) are scored by Avg@32 at $\tau=1.0$ to reduce variance, while the larger benchmarks use greedy single-sample evaluation. The reported average is the unweighted mean over the four benchmarks where DAPO and DAPO + RSP differ at peak (MATH-500, AIME 2024, GSM8K, Minerva Math); the remaining three are tied (Table 5 caption).

336 **Results.** Figure 5 shows the seven-benchmark average accuracy across training steps, and Table 5
 337 reports per-benchmark accuracy at each method’s peak step.

Table 5: Per-benchmark accuracy (%) at each method’s peak step on Qwen2.5-Math-7B. **Bold** denotes the better of the two RL methods on each benchmark. Avg is the unweighted four-benchmark mean.

Method	GSM8K	MATH-500	Minerva	AIME 24	Avg
Base	57.2	52.6	13.6	8.6	33.00
DAPO	86.3	78.2	37.5	22.6	56.15
DAPO + RSP	87.7	78.6	39.7	23.6	57.40

338 **Interpretation.** Two patterns emerge. First, DAPO
 339 alone peaks at step 80 (56.15%) and then declines, reach-
 340 ing 52.82% by step 100; DAPO + RSP keeps climbing
 341 to a 57.40% peak at step 90 and stays at 56.78% by
 342 step 100. This is consistent with input-space resam-
 343 pling acting as a *complementary* source of diversity to
 344 DAPO’s loss-level entropy management: RSP injects
 345 independent perturbations across rollouts and does not
 346 depend on the policy’s own entropy. Second, the peak
 347 itself improves (+1.25 pp), so RSP both *delays collapse*
 348 and *raises the attainable performance*. Both patterns
 349 are facets of the same mechanism: RSP’s early-stage
 350 branch opening (Section 3.4), the classical exploration-
 351 exploitation pattern in RL [Sutton and Barto, 2018],
 352 slows reward growth at first (the two curves cross around
 353 step 20) but compounds into both delayed collapse and
 354 a higher peak. We treat this as preliminary evidence that input-space and loss-level diversity mecha-
 355 nisms compose; limitations and broader robustness checks are discussed in Appendix F.6.

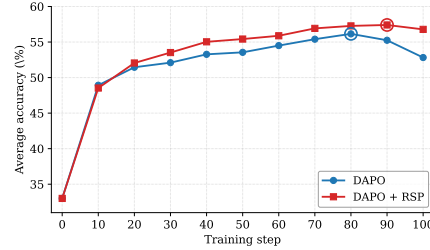


Figure 5: Seven-benchmark average accuracy on Qwen2.5-Math-7B. DAPO + RSP reaches a higher peak (step 90) and stays stable through step 100, while DAPO declines after step 80.

356 6 Conclusion

357 We studied Random Soft Prompts (RSPs), isotropic Gaussian vectors fitted to the entrywise mean
 358 and variance of the pretrained embedding table and injected without training. Theoretically, the RSP
 359 contribution at each attention layer is exactly captured by a scalar attention mass whose fixed-gap
 360 upper envelope shrinks under cache growth, and under a local affine analysis RSPs can reach top-
 361 K entry regions that rank-preserving temperature scaling cannot create (§3). Empirically, RSPs
 362 raise early-stage entropy while later generation stabilizes, and the accumulation gain disappears
 363 when one RSP is shared across samples (§4). Because RSP is rank-changing while temperature
 364 is rank-preserving, the two are complementary, suggesting a sampling-level remedy for entropy
 365 collapse in GRPO [Shao et al., 2024] beyond DAPO [Yu et al., 2025] and CE-GPPO [Su et al., 2026];
 366 training-time validation, extension beyond mathematical reasoning, and a principled choice of RSP
 367 length and injection position—together with the task-side bound $p_{\min}(x) > 0$ of Proposition 2—are
 368 natural next steps.

369 References

- 370 Farhan Ahmed, Yuya Jeremy Ong, and Chad DeLuca. LogitScope: A Framework for Analyzing
 371 LLM Uncertainty through Information Metrics. In *ICLR Workshop*, 2026.
- 372 Nora Belrose, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella
 373 Biderman, and Jacob Steinhardt. Eliciting Latent Predictions from Transformers with the Tuned
 374 Lens. *arXiv:2303.08112*, 2023.
- 375 Jianlyu Chen, Shitao Xiao, Peitian Zhang, Kun Luo, Defu Lian, and Zheng Liu. M3-
 376 Embedding: Multi-Linguality, Multi-Functionality, Multi-Granularity Text Embeddings Through
 377 Self-Knowledge Distillation. In *Findings of ACL*, 2024.

378 Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared
379 Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, et al. Evaluating Large
380 Language Models Trained on Code. *arXiv:2107.03374*, 2021.

381 Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser,
382 Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John
383 Schulman. Training Verifiers to Solve Math Word Problems. *arXiv:2110.14168*, 2021.

384 Jeremy M. Cohen, Elan Rosenfeld, and J. Zico Kolter. Certified Adversarial Robustness via Random-
385 ized Smoothing. In *ICML*, 2019.

386 Martin Ester, Hans-Peter Kriegel, Jörg Sander, and Xiaowei Xu. A Density-Based Algorithm for
387 Discovering Clusters in Large Spatial Databases with Noise. In *KDD*, 1996.

388 Meire Fortunato, Mohammad Gheshlaghi Azar, Bilal Piot, Jacob Menick, Ian Osband, Alex Graves,
389 Vlad Mnih, Remi Munos, Demis Hassabis, Olivier Pietquin, Charles Blundell, and Shane Legg.
390 Noisy Networks for Exploration. In *ICLR*, 2018.

391 Mor Geva, Roei Schuster, Jonathan Berant, and Omer Levy. Transformer Feed-Forward Layers Are
392 Key-Value Memories. In *EMNLP*, 2021.

393 Sachin Goyal, Ziwei Ji, Ankit Singh Rawat, Aditya Krishna Menon, Sanjiv Kumar, and Vaishnavh
394 Nagarajan. Think before You Speak: Training Language Models with Pause Tokens. In *ICLR*,
395 2024.

396 Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad
397 Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela
398 Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem
399 Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava
400 Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya
401 Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang
402 Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song,
403 Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan,
404 Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina
405 Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang,
406 Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire
407 Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron,
408 Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang,
409 Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer
410 van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang,
411 Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua
412 Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani,
413 Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz
414 Malik, Kuenley Chiu, Kunal Bhatta, Kushal Lakhotia, Lauren Rantala-Yearly, Laurens van der
415 Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo,
416 Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat
417 Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya
418 Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman
419 Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang,
420 Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic,
421 Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu,
422 Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira
423 Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain
424 Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar
425 Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov,
426 Shaoliang Nie, Sharan Narang, Sharath Raparthy, Sheng Shen, Shengye Wan, Shruti Bhosale,
427 Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane
428 Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha,
429 Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal
430 Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gouget,
431 Virginie Do, Vish Vogeti, Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin

432 Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide
 433 Xia, Xinfeng Xie, Xuchao Jia, Xuwei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei,
 434 Yi Wen, Yiwen Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan,
 435 Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey,
 436 Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma,
 437 Alex Boesenber, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo,
 438 Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew
 439 Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita
 440 Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh
 441 Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola,
 442 Bhargavi Paranjape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence,
 443 Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu,
 444 Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris
 445 Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel
 446 Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich,
 447 Didem Foss, Dingkan Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine
 448 Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban
 449 Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippas Kokkinos, Firat
 450 Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella
 451 Schwarz, Gada Badeer, Georgia Swee, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi Zhang,
 452 Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha,
 453 Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan
 454 Zhan, Ibrahim Damlaj, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai
 455 Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya,
 456 Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica
 457 Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan
 458 Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal,
 459 Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelena, Keqian Li, Kiran
 460 Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A,
 461 Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca
 462 Wehrstedt, Madian Khabza, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson,
 463 Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally,
 464 Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov,
 465 Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat,
 466 Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White,
 467 Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich
 468 Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem
 469 Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager,
 470 Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang,
 471 Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra,
 472 Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ
 473 Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh,
 474 Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji
 475 Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Lindsay, Sheng Feng, Shenghao Lin,
 476 Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang,
 477 Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe,
 478 Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny
 479 Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara
 480 Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou,
 481 Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish
 482 Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov,
 483 Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian
 484 Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi,
 485 Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu Wang, Yu Zhao, Yuchen Hao,
 486 Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu
 487 Yang, Zhiwei Zhao, and Zhiyu Ma. The Llama 3 Herd of Models. *arXiv:2407.21783*, 2024.

488 Shibo Hao, Sainbayar Sukhbaatar, DiJia Su, Xian Li, Zhiting Hu, Jason Weston, and Yuandong Tian.
 489 Training Large Language Models to Reason in a Continuous Latent Space. In *ICLR*, 2025.

490 Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu,
491 Xu Han, Yujie Huang, Yankai Zhang, Jie Liu, Lei Qi, Zhiyuan Liu, and Maosong Sun. Olympiad-
492 Bench: A Challenging Benchmark for Promoting AGI with Olympiad-Level Bilingual Multimodal
493 Scientific Problems. In *Findings of ACL*, 2024.

494 Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea
495 Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-Efficient Transfer Learning for NLP.
496 In *ICML*, 2019.

497 Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang,
498 and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models. In *ICLR*, 2022.

499 Neel Jain, Ping-yeh Chiang, Yuxin Wen, John Kirchenbauer, Hong-Min Chu, Gowthami Somepalli,
500 Brian R. Bartoldson, Bhavya Kailkhura, Avi Schwarzschild, Aniruddha Saha, Micah Goldblum,
501 Jonas Geiping, and Tom Goldstein. NEFTune: Noisy Embeddings Improve Instruction Finetuning.
502 In *ICLR*, 2024.

503 Xinyue Kang, Diwei Shi, and Li Chen. Model Whisper: Steering Vectors Unlock Large Language
504 Models’ Potential in Test-time. In *AAAI*, 2026.

505 Brian Lester, Rami Al-Rfou, and Noah Constant. The Power of Scale for Parameter-Efficient Prompt
506 Tuning. In *EMNLP*, 2021.

507 Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay
508 Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, Yuhuai Wu, Behnam
509 Neyshabur, Guy Gur-Ari, and Vedant Misra. Solving Quantitative Reasoning Problems with
510 Language Models. *arXiv:2206.14858*, 2022.

511 Xiang Lisa Li and Percy Liang. Prefix-Tuning: Optimizing Continuous Prompts for Generation. In
512 *ACL-IJCNLP*, 2021.

513 Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike,
514 John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s Verify Step by Step. In *ICLR*, 2024.

515 Xiao Liu, Yanan Zheng, Zhengxiao Du, Ming Ding, Yujie Qian, Zhilin Yang, and Jie Tang. GPT
516 Understands, Too. *AI Open*, 5:208–215, 2024.

517 Aman Madaan and Amir Yazdanbakhsh. Text and Patterns: For Effective Chain of Thought, It Takes
518 Two to Tango. *arXiv:2209.07686*, 2022.

519 Alexander Robey, Eric Wong, Hamed Hassani, and George J. Pappas. SmoothLLM: Defending Large
520 Language Models Against Jailbreaking Attacks. *Transactions on Machine Learning Research*,
521 2025.

522 Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang,
523 Mingchuan Zhang, Y.K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the Limits of
524 Mathematical Reasoning in Open Language Models. *arXiv:2402.03300*, 2024.

525 Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin Wu, Wang Zhang, Ru Zhang, Yanghua
526 Peng, Haibin Lin, and Chuan Wu. HybridFlow: A Flexible and Efficient RLHF Framework.
527 *arXiv:2409.19256*, 2024.

528 Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov.
529 Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *Journal of Machine
530 Learning Research*, 15:1929–1958, 2014.

531 Zhenpeng Su, Leiyu Pan, Minxuan Lv, Yuntao Li, Wenping Hu, Fuzheng Zhang, Kun Gai, and Guorui
532 Zhou. CE-GPPO: Coordinating Entropy via Gradient-Preserving Clipping Policy Optimization in
533 Reinforcement Learning. In *ACL*, 2026.

534 Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2
535 edition, 2018.

- 536 Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shixuan Liu, Rui Lu, Kai Dang, Xionghui Chen,
537 Jianxin Yang, Zhenru Zhang, Yuqiong Liu, An Yang, Andrew Zhao, Yang Yue, Shiji Song, Bowen
538 Yu, Gao Huang, and Junyang Lin. Beyond the 80/20 Rule: High-Entropy Minority Tokens Drive
539 Effective Reinforcement Learning for LLM Reasoning. In *NeurIPS*, 2025.
- 540 Yige Xu, Xu Guo, Zhiwei Zeng, and Chunyan Miao. SoftCoT: Soft Chain-of-Thought for Efficient
541 Reasoning with LLMs. In *ACL*, 2025.
- 542 An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao, Bowen Yu, Chengpeng Li, Dayiheng Liu,
543 Jianhong Tu, Jingren Zhou, Junyang Lin, Keming Lu, Mingfeng Xue, Runji Lin, Tianyu Liu,
544 Xingzhang Ren, and Zhenru Zhang. Qwen2.5-Math Technical Report: Toward Mathematical
545 Expert Model via Self-Improvement. *arXiv:2409.12122*, 2024.
- 546 Wengao Ye, Yan Liang, and Lianlei Shan. Thinking on the Fly: Test-Time Reasoning Enhancement
547 via Latent Thought Policy Optimization. In *ICLR*, 2026.
- 548 Qiying Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian
549 Fan, Gaohong Liu, Juncai Liu, Lingjun Liu, Xin Liu, Haibin Lin, Zhiqi Lin, Bole Ma, Guangming
550 Sheng, Yuxuan Tong, Chi Zhang, Mofan Zhang, Ru Zhang, Wang Zhang, Hang Zhu, Jinhua Zhu,
551 Jiaze Chen, Jiangjie Chen, Chengyi Wang, Hongli Yu, Yuxuan Song, Xiangpeng Wei, Hao Zhou,
552 Jingjing Liu, Wei-Ying Ma, Ya-Qin Zhang, Lin Yan, Yonghui Wu, and Mingxuan Wang. DAPO:
553 An Open-Source LLM Reinforcement Learning System at Scale. In *NeurIPS*, 2025.
- 554 Weihao Zeng, Yuzhen Huang, Qian Liu, Wei Liu, Keqing He, Zejun Ma, and Junxian He. SimpleRL-
555 Zoo: Investigating and Taming Zero Reinforcement Learning for Open Base Models in the Wild.
556 *arXiv:2503.18892*, 2025.
- 557 Guibin Zhang, Muxin Fu, and Shuicheng Yan. MemGen: Weaving Generative Latent Memory for
558 Self-Evolving Agents. In *ICLR*, 2026.

559 A Experimental Setup and Full Accuracy Breakdown

560 Table 6 provides the full per-position accuracy breakdown (Prefix, Infix, Suffix) corresponding to
561 the best-across-positions results summarized in Table 1 in the main text. Table 7 lists the number of
562 RSP tokens L used for each model–benchmark pair, selected from $\{10, 15, 20\}$. All experiments are
563 conducted on a single NVIDIA A6000 40GB GPU with greedy decoding, a maximum generation
564 length of 3,072 tokens for MATH-500 and GSM8K, and 4,096 tokens for AIME24. Batch size is
565 fixed per model (16 for LLaMA-3.1-8B-Instruct and Qwen2.5-Math-7B variants, 32 for Qwen2.5-
566 Math-1.5B variants).

Table 6: Accuracy (%) across model configurations, injection positions, and benchmarks. Values in parentheses indicate the difference from the baseline. **Bold** denotes the best result and underline denotes the second best for each model–benchmark pair.

Model	Mode	MATH-500	GSM8K	AIME24
<i>Instruct Models</i>				
LLaMA-3.1-8B-Instruct	Baseline	52.20	85.75	<u>6.67</u>
	Prefix	45.00 (−7.2)	76.35 (−9.4)	3.33 (−3.3)
	Infix	49.40 (−2.8)	85.97 (+0.2)	10.00 (+3.3)
	Suffix	<u>49.40</u> (−2.8)	86.73 (+1.0)	10.00 (+3.3)
Qwen2.5-Math-7B-Instruct	Baseline	83.20	95.45	<u>13.33</u>
	Prefix	81.40 (−1.8)	94.92 (−0.5)	13.33 (+0.0)
	Infix	84.20 (+1.0)	95.60 (+0.2)	16.67 (+3.3)
	Suffix	<u>83.40</u> (+0.2)	95.68 (+0.2)	16.67 (+3.3)
Qwen2.5-Math-1.5B-Instruct	Baseline	73.00	<u>85.29</u>	10.00
	Prefix	74.80 (+1.8)	85.29 (+0.0)	<u>13.33</u> (+3.3)
	Infix	75.20 (+2.2)	85.75 (+0.5)	6.67 (−3.3)
	Suffix	74.20 (+1.2)	84.69 (−0.6)	20.00 (+10.0)
<i>Base Models (Raw Text)</i>				
Qwen2.5-Math-7B	Baseline	72.20	84.15	6.67
	Prefix	<u>71.60</u> (−0.6)	84.31 (+0.2)	16.67 (+10.0)
	Infix	63.20 (−9.0)	64.29 (−19.9)	16.67 (+10.0)
	Suffix	55.60 (−16.6)	54.13 (−30.0)	<u>13.33</u> (+6.7)
Qwen2.5-Math-1.5B	Baseline	<u>61.40</u>	77.86	16.67
	Prefix	64.40 (+3.0)	74.30 (−3.6)	10.00 (−6.7)
	Infix	63.20 (+1.8)	73.92 (−3.9)	<u>10.00</u> (−6.7)
	Suffix	54.60 (−6.8)	58.61 (−19.3)	3.33 (−13.3)
<i>Base Models + ChatML (Format Mismatch)</i>				
Qwen2.5-Math-7B	Baseline	52.20	58.30	23.33
	Prefix	59.20 (+7.0)	56.41 (−1.9)	<u>3.33</u> (−20.0)
	Infix	<u>62.00</u> (+9.8)	69.07 (+10.8)	23.33 (+0.0)
	Suffix	70.40 (+18.2)	75.97 (+17.7)	23.33 (+0.0)
Qwen2.5-Math-1.5B	Baseline	34.40	37.45	<u>13.33</u>
	Prefix	63.40 (+29.0)	67.02 (+29.6)	20.00 (+6.7)
	Infix	39.20 (+4.8)	50.42 (+13.0)	6.67 (−6.7)
	Suffix	<u>51.00</u> (+16.6)	<u>50.64</u> (+13.2)	<u>13.33</u> (+0.0)

Table 7: Number of RSP tokens (L) used for each model and benchmark.

Model	MATH-500	GSM8K	AIME24
LLaMA-3.1-8B-Instruct	15	20	15
Qwen2.5-Math-7B-Instruct	20	20	20
Qwen2.5-Math-1.5B-Instruct	20	10	20
Qwen2.5-Math-7B	10	10	20
Qwen2.5-Math-1.5B	10	10	20
Qwen2.5-Math-1.5B (ChatML)	20	20	10
Qwen2.5-Math-7B (ChatML)	20	20	20

567 A.1 RSP Injection Positions and Prompt Templates

568 Below we show the prompt structure for each injection position across all model types. Blue text
569 indicates RSP embeddings, and purple text indicates special tokens.

570 A.1.1 Prefix

571 RSP embeddings are concatenated before the prompt embeddings. No text modification is applied.

572 ChatML (Qwen Instruct / Base + ChatML).

```
[Random Embeddings (L tokens)]
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{<|im_end|>
<|im_start|>user
{question}<|im_end|>
<|im_start|>assistant
```

574 LLaMA Chat Template.

```
[Random Embeddings (L tokens)]
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
<|start_header_id|>user<|end_header_id|>
{question}<|eot_id|>
<|start_header_id|>assistant<|end_header_id|>
```

576 Raw Text (Qwen Base).

```
[Random Embeddings (L tokens)]
{question}
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
```

578 A.1.2 Infix

579 A description text and special tokens are inserted into the prompt. The special token positions are
580 then replaced with RSP embeddings at the embedding level.

581 ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{<|im_end|>
<|im_start|>user
{question}

There are {L} special tokens that contain compressed latent reasoning information that
might be useful for your reasoning. If these tokens are useful for your case, you can
use them as reference. If these tokens are not useful for your case, you can ignore
them and focus back to solving the problem.

Here are the {L} special tokens: <special_token_1>...<special_token_L>
<|im_end|>
<|im_start|>assistant
```

583 LLaMA Chat Template.

```
<|begin_of_text|>
<|start_header_id|>system<|end_header_id|>
Cutting Knowledge Date: December 2023
Today Date: 26 Jul 2024
Please reason step by step, and put your final answer within \boxed{<|eot_id|>
<|start_header_id|>user<|end_header_id|>
```

584

{question}

There are $\{L\}$ special tokens that contain compressed latent reasoning information that might be useful for your reasoning. If these tokens are useful for your case, you can use them as reference. If these tokens are not useful for your case, you can ignore them and focus back to solving the problem.

Here are the $\{L\}$ special tokens:

```
<reserved_special_token_0>...  
<reserved_special_token_L>  
<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

585

586 Raw Text (Qwen Base).

{question}

There are $\{L\}$ special tokens that contain compressed latent reasoning information that might be useful for your reasoning. If these tokens are useful for your case, you can use them as reference. If these tokens are not useful for your case, you can ignore them and focus back to solving the problem.

Here are the $\{L\}$ special tokens: $\langle \text{special_token_1} \rangle \dots \langle \text{special_token_L} \rangle$

Please reason step by step, and put your final answer within `\boxed{\}`.

587

588 A.1.3 Suffix

589 For chat-templated models, RSP embeddings are inserted immediately before the end-of-turn token.
590 For raw text models, RSP embeddings are concatenated at the end of the prompt.

591 ChatML (Qwen Instruct / Base + ChatML).

```
<|im_start|>system  
Please reason step by step, and put your final answer within \boxed{<|im_end|>  
<|im_start|>user  
{question}[Random Embeddings (L tokens)]<|im_end|>  
<|im_start|>assistant
```

592

593 LLaMA Chat Template.

```
<|begin_of_text|>  
<|start_header_id|>system<|end_header_id|>  
Cutting Knowledge Date: December 2023  
Today Date: 26 Jul 2024  
Please reason step by step, and put your final answer within \boxed{<|eot_id|>  
<|start_header_id|>user<|end_header_id|>  
{question}[Random Embeddings (L tokens)]<|eot_id|>  
<|start_header_id|>assistant<|end_header_id|>
```

594

595 Raw Text (Qwen Base).

```
{question}  
Please reason step by step, and put your final answer within \boxed{<|eot_id|>[Random  
Embeddings (L tokens)]
```

596

597 A.2 Answer Extraction and Grading

598 The final answer is extracted from the model output through a fallback chain. Each step is attempted
599 in order; if extraction fails, the next step is tried.

Answer Extraction Fallback Chain

1. "final answer is \$...\$. I hope" pattern (Minerva format)
2. `\boxed{...}`: last non-empty box is used; empty `\boxed{}` are skipped
3. Text following "the answer is"
4. Text following "final answer is"
5. Last number in the output (regex fallback)

Extracted answers are normalized by removing \$, \left, \right, and unit strings. Grading is performed via symbolic equivalence checking: exact string match, numerical comparison (`math.isclose, rel_tol=1e-4`), and SymPy symbolic simplification.

A.3 Reproduced Prior Work Hyperparameters

All prior methods are reproduced on Qwen2.5-Math-1.5B-Instruct and Qwen2.5-Math-7B-Instruct, evaluated with greedy decoding (temperature = 0) and our unified answer extraction pipeline.

TTSV. Prefix length 20, trained for 20 epochs with AdamW optimizer, batch size 2, gradient accumulation 8 steps, and linear learning rate schedule. Learning rate is 1e-3 for Qwen-1.5B and 1e-5 for Qwen-7B.

LTPO. Table 8 reports the per-benchmark hyperparameters.

Table 8: LTPO hyperparameters.

Parameter	GSM8K	MATH-500	AIME24
tokens	8	8	8
steps	20	20	20
top_k	10	10	10
sigma	20	20	5
sigma_decay	0.95	0.95	0.95
lr	1e-2	5e-3	5e-2

SoftCoT. Projection module trained for 10 epochs. Thought tokens: 32 during training, 4 during evaluation. Batch size 4 for GSM8K, 1 for MATH with gradient accumulation 4. The small (assistant) model is Qwen2.5-Math-1.5B-Instruct in both configurations, while the large model is Qwen2.5-Math-1.5B-Instruct (learning rate 2e-5) or Qwen2.5-Math-7B-Instruct (learning rate 5e-6). For MATH-500 and AIME24 evaluation, we use the projection module trained on GSM8K, as it yields higher accuracy than the one trained on MATH.

A.4 Full Entropy Curves

Figure 6 shows the full normalized (0–100%) generation trajectories for entropy, top-1 probability, and varentropy, and Table 9 reports the corresponding scalar statistics including overall means, first-5% means, and the average number of generated tokens per problem. All methods converge to similar levels after the initial generation stage, confirming that the distributional changes induced by RSP are localized to the early reasoning phase.

Table 9: Per-step statistics for Qwen2.5-Math-1.5B-Instruct on MATH-500 across the full generation and the first 5% of generation steps, together with the average number of generated tokens per problem. Top-1 Prob (overall) is reported as a weighted average over the first-10%/middle-last-10% segment means with weights 0.1/0.8/0.1. Extension of Figure 3 (main text) and Figure 6.

Metric	Baseline	Prefix	Infix	Suffix	LTPO	SoftCoT	TTSV
Tokens (mean)	575.7	558.4	549.9	573.3	592.3	594.4	577.4
Entropy (first 5%)	0.1332	0.1366	0.1402	0.1941	0.1662	0.4218	0.1359
Entropy (overall)	0.0871	0.0881	0.0899	0.1049	0.0909	0.1059	0.0864
Top-1 Prob (first 5%)	0.9535	0.9523	0.9525	0.9373	0.9437	0.9156	0.9534
Top-1 Prob (overall)	0.9684	0.9681	0.9678	0.9647	0.9683	0.9651	0.9685
Varentropy (first 5%)	0.2181	0.2207	0.2463	0.4010	0.2953	2.2234	0.2174
Varentropy (overall)	0.1353	0.1375	0.1422	0.2038	0.1452	0.2404	0.1361

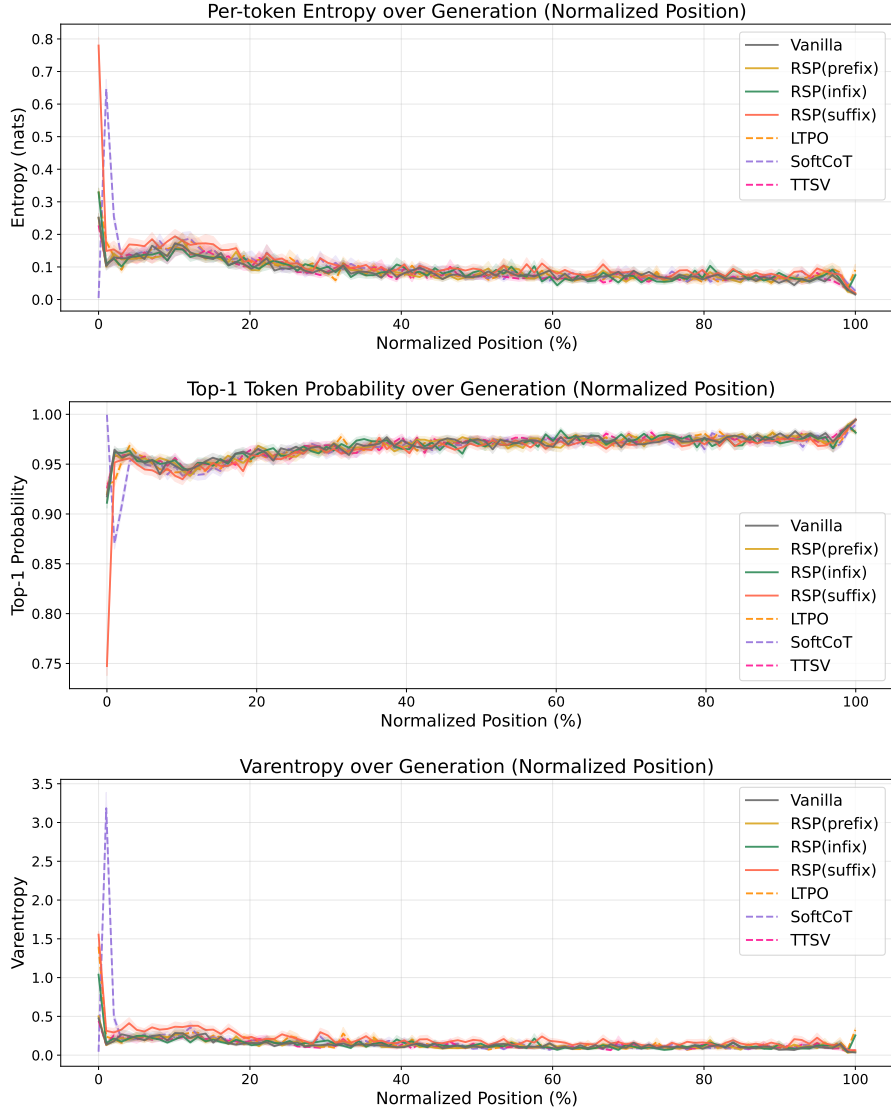


Figure 6: Mean entropy (top), top-1 probability (middle), and varentropy (bottom) over the full normalized generation trajectory (0–100%). Averaged over MATH-500, shading indicates ± 1 SEM. Solid lines: RSP variants and the baseline. Dashed lines: existing soft prompt methods. All methods converge after the initial stage.

623 B Token-Level Distribution Metrics

624 This appendix provides formal definitions for the metrics used in Section 4.4. Numerical results
 625 are reported in Table 3 of the main text. We extract first-token logits via a single forward pass (no
 626 autoregressive generation) and store the top-100 token ids and logit values per problem. Temperature
 627 conditions reuse the baseline logits scaled by $1/\tau$, requiring no separate inference. RSP draws 16
 628 independent seeds per problem (MATH-500), and reported metrics are the mean across the 16 seeds.
 629 Pass@1 accuracy is computed over 16 samples per problem, with the baseline at $\tau=1.0$ reaching
 630 $73.92 \pm 0.78\%$. Let P_{base} denote the baseline next-token distribution at $\tau = 1.0$ and P_{target} the
 631 candidate distribution being compared (e.g., baselines at $\tau=2.0, 3.0$ or RSP at $\tau = 1.0$). All metrics
 632 are computed analytically from saved logits at the first generation step.

633 **Spearman rank correlation.** Spearman’s ρ is the Pearson correlation coefficient between the token
 634 ranks of two distributions:

$$\rho = \text{Pearson}(\text{rank}(P_{\text{target}}), \text{rank}(P_{\text{base}})). \quad (4)$$

635 Because temperature scaling $P^{(\tau)} \propto P^{1/\tau}$ is a strictly monotone transform, the rank order of tokens
 636 is preserved exactly, so $\rho = 1$ for any $\tau > 0$ as a mathematical identity. Any value $\rho < 1$ therefore
 637 indicates rank reorganization beyond what temperature scaling can produce.

638 **Mass outside top- K .** Let $\text{TopK}(P_{\text{base}})$ be the set of K tokens with the highest probability under
 639 P_{base} . The probability mass that the candidate distribution places *outside* this set is

$$\text{MassOutside}_K = 1 - \sum_{t \in \text{TopK}(P_{\text{base}})} P_{\text{target}}(t). \quad (5)$$

640 This directly measures support expansion: how much probability the candidate assigns to tokens that
 641 are not in the baseline’s preferred set. Reported values use $K = 10$.

642 **Jensen–Shannon divergence.** The symmetrized, bounded version of Kullback–Leibler divergence:

$$\text{JS}(P, Q) = \frac{1}{2} \text{KL}(P \parallel M) + \frac{1}{2} \text{KL}(Q \parallel M), \quad M = \frac{1}{2}(P + Q). \quad (6)$$

643 We use base-2 logarithms so that $\text{JS} \in [0, 1]$. Tokens absent from the stored top-100 are assigned a
 644 sentinel logit (-10^9) before softmax, which is negligible for our setting.

C Prompt-Law Properties Used by the Theory

This appendix isolates the only properties of the RSP law used in the main text: centeredness, direction-agnostic covariance under a variance budget, and positive probability on every nonempty open set. We avoid stronger semantic claims because Section 3.4 needs only these geometric and measure-theoretic facts.

C.1 Centered Perturbations Do Not Impose Systematic First-Order Drift

Let Δ_{ij} be the baseline logit gap between tokens i and j , and write the first-order surrogate as

$$\Delta_{ij}(\bar{h}) = \Delta_{ij} + b_{ij}^\top \bar{h}.$$

Proposition 3 (No systematic first-order drift). *If $\mathbb{E}[\bar{h}] = 0$, then*

$$\mathbb{E}[\Delta_{ij}(\bar{h})] = \Delta_{ij}$$

for every token pair (i, j) . If a deterministic prompt h_0 is reused across runs, then

$$\Delta_{ij}(h_0) - \Delta_{ij} = b_{ij}^\top h_0$$

is a fixed directional bias.

Proof. The centered case is immediate from linearity of expectation:

$$\mathbb{E}[\Delta_{ij}(\bar{h})] = \Delta_{ij} + b_{ij}^\top \mathbb{E}[\bar{h}] = \Delta_{ij}.$$

The deterministic case follows by substitution. $\square$

This proposition rules out a run-averaged first-order bias. It does not say that individual prompt draws leave the logits unchanged.

C.2 Proof of Theorem 2

For a unit vector b , the first-order perturbation energy available along b is $\text{Var}_{h \sim D}(b^\top h) = b^\top \Sigma_D b$. The worst-case directional energy is therefore the minimum Rayleigh quotient of Σ_D .

Proof. For any symmetric covariance matrix Σ_D ,

$$\min_{\|b\|=1} b^\top \Sigma_D b = \lambda_{\min}(\Sigma_D).$$

Let the eigenvalues of Σ_D be $\lambda_1, \dots, \lambda_d$. Then

$$\lambda_{\min}(\Sigma_D) \leq \frac{1}{d} \sum_{m=1}^d \lambda_m = \frac{\text{tr}(\Sigma_D)}{d} \leq \frac{\rho^2}{d}.$$

Equality holds if and only if all eigenvalues are equal, i.e., $\Sigma_D = (\rho^2/d)\mathbf{I}_d$. $\square$

The proof uses only covariance. We choose a Gaussian law for RSP because it adds the open-set reachability property proved next.

C.3 Open-Set Reachability of Isotropic Gaussian Prompts

Proposition 4 (Open-set reachability). *Let $\bar{h} \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_d)$ with $\sigma > 0$. Then every nonempty open set $U \subseteq \mathbb{R}^d$ satisfies*

$$\Pr(\bar{h} \in U) > 0.$$

Proof. The Gaussian density

$$(2\pi\sigma^2)^{-d/2} \exp\left(-\frac{\|\bar{h}\|^2}{2\sigma^2}\right)$$

is strictly positive at every point of $\mathbb{R}^d$. Every nonempty open set contains a ball of positive Lebesgue measure, and integrating a strictly positive density over that ball gives positive probability. $\square$

This is the only support property used in Proposition 1.

674 D Proofs for the Transformer-Level Mechanism

675 This appendix follows the same order as the main theory section: exploration, annealing, and
676 performance gain. The prompt-law facts used before these proofs are collected in Appendix C.

677 D.1 Exploration: attention decomposition and perturbation size

678 **Proposition 5** (Attention decomposition). *For one attention head at query position i in layer ℓ , let*
679 *$\alpha_{ij}^{(\ell)}$ be the attention weights over $n \geq 1$ unmasked real tokens and $L \geq 1$ random tokens, with finite*
680 *attention logits. Define*

$$w_{r,i}^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)}$$

681 and, for $j \in [n]$,

$$\tilde{\alpha}_{ij}^{(\ell)} := \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}}.$$

682 Then

$$\mathbf{o}_i^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)} + \boldsymbol{\eta}_i^{(\ell)},$$

683 where

$$\tilde{\mathbf{o}}_i^{(\ell)} := \sum_{j=1}^n \tilde{\alpha}_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} \quad \text{and} \quad \boldsymbol{\eta}_i^{(\ell)} := \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

684 *Proof.* The head output is

$$\mathbf{o}_i^{(\ell)} = \sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} + \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)}.$$

685 By the softmax positivity and the existence of at least one unmasked real token, $\sum_{j=1}^n \alpha_{ij}^{(\ell)} =$
686 $1 - w_{r,i}^{(\ell)} > 0$, so

$$\sum_{j=1}^n \alpha_{ij}^{(\ell)} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \sum_{j=1}^n \frac{\alpha_{ij}^{(\ell)}}{1 - w_{r,i}^{(\ell)}} \mathbf{v}_j^{(\ell)} = (1 - w_{r,i}^{(\ell)}) \tilde{\mathbf{o}}_i^{(\ell)}.$$

687 Substituting this identity gives the decomposition. □

688 **Corollary 1** (Magnitude scales with random-token attention). *For every realization of the random*
689 *values,*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

690 *Proof.* By the triangle inequality,

$$\|\boldsymbol{\eta}_i^{(\ell)}\| = \left\| \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \mathbf{v}_{r_j}^{(\ell)} \right\| \leq \sum_{j=1}^L \alpha_{i,n+j}^{(\ell)} \|\mathbf{v}_{r_j}^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\|.$$

691 □

692 Corollary 1 is the only quantitative fact needed in the main text. No independence assumption
693 between attention weights and random keys is required. Once $w_{r,i}^{(\ell)}$ is small, the random contribution
694 must be small as well.

695 D.2 Corollaries of Theorem 1 on cache growth

696 The fixed-gap decay bound yields the corollaries below.

697 **Corollary 2** (Sequence-wise annealing of the fixed-gap envelope). *For fixed L and $\Delta_i^{(\ell)}$, the upper*
 698 *bound*

$$f(n) = \frac{L}{L + n \exp(\Delta_i^{(\ell)} / \sqrt{d})}$$

699 *is strictly decreasing in n and tends to 0 as $n \rightarrow \infty$. Cache growth alone therefore narrows the*
 700 *largest RSP attention mass compatible with that gap.*

701 During autoregressive decoding $\Delta_i^{(\ell)}$ also moves because queries, keys, and hidden states co-evolve.
 702 The accurate reading is not “the realized mass decreases at every step” but “the envelope compatible
 703 with a fixed gap shrinks as the cache grows.”

704 *Proof.* Differentiate $f(n)$ with respect to n :

$$f'(n) = -\frac{L \exp(\Delta_i^{(\ell)} / \sqrt{d})}{\left(L + n \exp(\Delta_i^{(\ell)} / \sqrt{d})\right)^2} < 0.$$

705 Thus $f(n)$ is strictly decreasing, and its denominator diverges linearly in n while its numerator is
 706 fixed, so $f(n) \rightarrow 0$. $\square$

707 **Corollary 3** (Vanishing random contribution under annealing). *If $n \rightarrow \infty$ while L and $\Delta_i^{(\ell)}$ remain*
 708 *fixed, $w_{r,i}^{(\ell)} \rightarrow 0$, and for every realization of the random values*

$$\|\boldsymbol{\eta}_i^{(\ell)}\| \leq w_{r,i}^{(\ell)} \max_{j \in [L]} \|\mathbf{v}_{r_j}^{(\ell)}\| \rightarrow 0.$$

709 *Proof.* $w_{r,i}^{(\ell)} \rightarrow 0$ by Corollary 2. Apply Corollary 1 to bound $\|\boldsymbol{\eta}_i^{(\ell)}\|$. The maximum norm is finite
 710 for any realization of finitely many sampled values. $\square$

711 D.3 Performance gain: from local admission to Pass@N

712 The main text §3.4 uses a first-order surrogate to state two claims: a baseline-excluded token can
 713 enter the local top- K set, and independent reruns can turn such local openings into Pass@ N gains.
 714 The proofs below use only one support condition on the centered prompt law D :

$$(S1) \quad D(U) > 0 \quad \text{for every nonempty open set } U \subseteq \mathbb{R}^d.$$

715 Proposition 4 shows that the isotropic Gaussian law of §3.4 satisfies (S1).

716 D.3.1 Autoregressive Formulation

717 In autoregressive decoding, the joint distribution over a sequence $y_{1:T}$ factorizes as

$$\Pr(y_{1:T} \mid x) = \prod_{t=1}^T p(y_t \mid x, y_{<t}),$$

718 where each step is governed by a prefix-conditioned distribution. Any perturbation introduced by
 719 RSP affects generation through a sequence of local changes to $p(y_t \mid x, y_{<t})$, rather than a single
 720 global distribution.

721 D.3.2 Local Linearization of Logits under RSP

722 Throughout this subsection, $\bar{h} \in \mathbb{R}^d$ denotes *one* centered prompt vector from the RSP definition
 723 (Eq. (1) in §3.1), treated as a per-token surrogate. The actual implementation uses a length- L prompt
 724 $\mathbf{H} = (h_1, \dots, h_L)$ with L independent draws; the arguments below isolate how one local perturbation

725 can shift the logits. We model the perturbed logits as a function $z_i(\bar{h})$, assuming local smoothness in
 726 a neighborhood of $\bar{h} = 0$:

$$z_i(\bar{h}) = z_i(0) + \nabla_h z_i(0)^\top \bar{h} + r_i(\bar{h}),$$

727 where $r_i(\bar{h}) = O(\|\bar{h}\|^2)$. Defining $a_i := \nabla_h z_i(0)$, we obtain the local affine surrogate

$$z_i^{\text{lin}}(\bar{h}) := z_i + a_i^\top \bar{h}.$$

728 This approximation is used as a first-order surrogate for branch opening, not as an exact global model
 729 of the transformer logits.

730 D.3.3 Proof of Proposition 1

731 Let

$$\mathcal{G}_{i,K} := \{\bar{h} \in \mathbb{R}^d : \#\{j \neq i : z_j^{\text{lin}}(\bar{h}) > z_i^{\text{lin}}(\bar{h})\} \leq K - 1\}.$$

732 *Proof.* Assume $\mathcal{G}_{i,K}$ contains a nonempty open set U . By (S1), $D(U) > 0$. Since $U \subseteq \mathcal{G}_{i,K}$,

$$\Pr_{\bar{h}}(\bar{h} \in \mathcal{G}_{i,K}) \geq D(U) > 0.$$

733 This is exactly the event that token i enters the surrogate top- K set.

734 A useful sufficient condition is the existence of some $\bar{h}_0$ at which token i strictly outranks all but
 735 at most $K - 1$ other tokens in the affine surrogate. By continuity, a neighborhood of $\bar{h}_0$ is then
 736 contained in $\mathcal{G}_{i,K}$. $\square$

737 D.3.4 Proof of Proposition 2

738 This proposition does not derive $p_{\min}(x) > 0$ from the mechanism. It only converts a task-side lower
 739 bound and independence into Pass@ N growth.

740 *Proof.* Let A_n be the event that the n -th independent RSP run reaches a correct trajectory on problem
 741 x , and write $p_n(x) := \Pr(A_n) \geq p_{\min}(x)$ as in (M1). Under (M2),

$$\Pr\left(\bigcap_{n=1}^N A_n^c\right) = \prod_{n=1}^N (1 - p_n(x)) \leq \prod_{n=1}^N (1 - p_{\min}(x)) = (1 - p_{\min}(x))^N.$$

742 Taking complements gives

$$\Pr(\text{Pass@}N \text{ on } x) = \Pr\left(\bigcup_{n=1}^N A_n\right) \geq 1 - (1 - p_{\min}(x))^N.$$

743 The bound $1 - x \leq e^{-x}$ for $x \in [0, 1]$ then yields

$$1 - (1 - p_{\min}(x))^N \geq 1 - e^{-N p_{\min}(x)}.$$

744 $\square$

E Attention Mass Analysis

This appendix formalizes the per-token attention mass reported in Figure 2 and the exclusion criteria applied to the reasoning span and the layer axis. For each of the 500 MATH-500 problems, a fresh RSP $\mathbf{H} \in \mathbb{R}^{L \times d}$ with $L = 10$ is sampled independently, so the reported heatmaps average over both problem variability and RSP-vector variability.

E.1 Per-Token Attention Mass

For each problem, we measure attention on the concatenated sequence [prompt; $\mathbf{H}$; generation], where the generation is the trajectory produced by the same model under matching RSP injection. Let $\alpha_{i,j}^{(\ell,h)}$ denote the post-softmax attention weight at layer ℓ , head h , from query position i to key position j , satisfying $\sum_j \alpha_{i,j}^{(\ell,h)} = 1$ under the causal mask. Let Q and R denote the index sets of question and RSP tokens with sizes $|Q|$ and $|R| = L$, and let H be the number of heads. The per-token attention mass that query i directs to each group at layer ℓ is

$$\text{PTM}_Q[\ell, i] = \frac{1}{|Q|H} \sum_{h=1}^H \sum_{j \in Q} \alpha_{i,j}^{(\ell,h)}, \quad \text{PTM}_R[\ell, i] = \frac{1}{|R|H} \sum_{h=1}^H \sum_{j \in R} \alpha_{i,j}^{(\ell,h)}. \quad (7)$$

Normalization by $|Q|$ and $|R|$ controls for group size, so both quantities express how much attention a single question (resp. RSP) token receives on average under the same softmax denominator. Question tokens thereby serve as a size-matched baseline that absorbs sequence-length effects common to both groups.

E.2 Heatmap Aggregation

We partition the layer indices and the reasoning position indices each into five equal-size bins $\{B_L^{(b)}\}_{b=1}^5$ and $\{B_P^{(b)}\}_{b=1}^5$. For sample s and target group $\bullet \in \{Q, R\}$, the value in cell (b_L, b_P) is

$$M_\bullet^{(s)}[b_L, b_P] = \frac{1}{|B_L^{(b_L)}| \cdot |B_P^{(b_P)}|} \sum_{\ell \in B_L^{(b_L)}} \sum_{i \in B_P^{(b_P)}} \text{PTM}_\bullet[\ell, i]. \quad (8)$$

Figure 2 reports the sample average of Eq. (8) over the 500 valid samples.

E.3 Excluding the `\boxed{\dots}` Span

The reasoning position range terminates strictly before the final `\boxed{\dots}` span. Chain-of-thought outputs decompose into a *text* (content) component and a *pattern* (template) component that contribute independently to downstream performance [Madaan and Yazdanbakhsh, 2022]. The `\boxed{\dots}` wrapper is a canonical pattern: a short, stereotyped span whose role is to complete the answer format rather than to extend reasoning. Including it would therefore inject a template-driven attention signature unrelated to reasoning dynamics.

E.4 Excluding the Final Two Layers

The layer axis further excludes the last two transformer layers. This choice is motivated by a standard late-layer effect rather than by any model-specific tuning decision.

Output-projection specialization. Late-layer hidden states lie in an approximately linear relationship with vocabulary logits and therefore function as a progressive linear readout rather than as representation-transforming blocks [Belrose et al., 2023]. Consistently, late-layer feed-forward modules have been characterized as output-distribution shaping mechanisms [Geva et al., 2021]. Attention in these layers therefore reflects output formation rather than reasoning computation.

Excluding the final two layers keeps the heatmap focused on reasoning-phase dynamics instead of readout-phase dynamics. This exclusion is not load-bearing for the main claim: the sequence-direction attenuation emphasized in the main text is also visible without it, and Theorem 1 concerns sequence-direction attenuation rather than layer-wise monotonicity.

784 E.5 Additional Suffix Heatmaps

785 Figure 7 reports the same per-token RSP attention analysis under suffix injection for Qwen2.5-Math-
 786 1.5B-Instruct and LLaMA-3.1-8B-Instruct. For the late-layer reasons discussed above, the pattern
 787 does not generalize uniformly across every cell; nevertheless, the overall trend is consistent across
 788 both models: question-token attention varies broadly across layers, whereas RSP attention shows the
 789 layer-wise and sequence-wise decay predicted by Theorem 1.

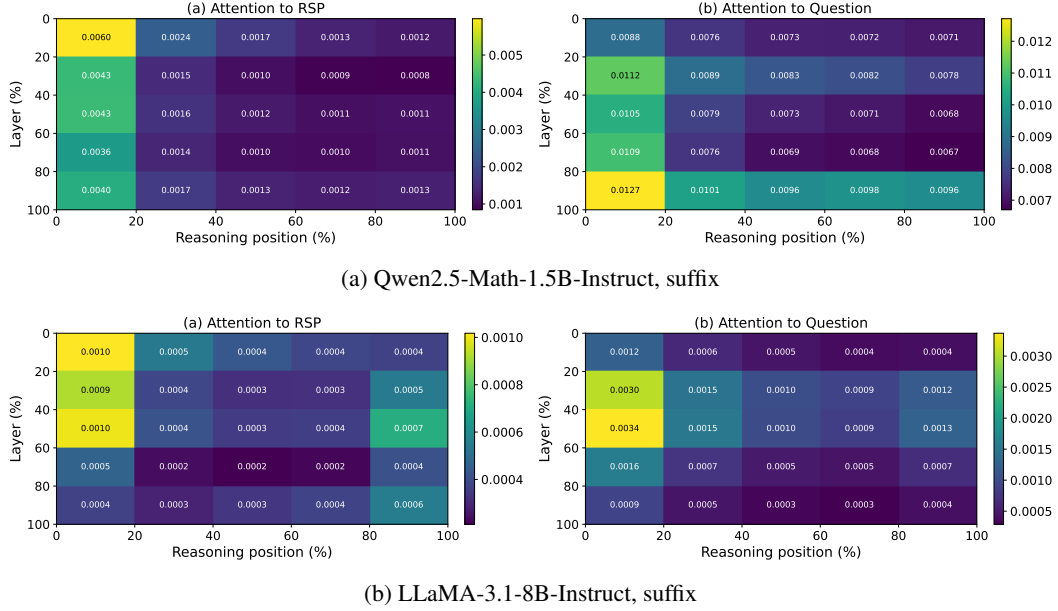


Figure 7: Per-token RSP attention mass under suffix injection for the remaining two models (500 MATH-500 samples each). Axes and preprocessing match Figure 2: x-axis is reasoning position binned into five quantiles, y-axis is layer depth binned into five quantiles (top = shallow), color encodes the 500-sample mean per-token attention assigned to RSP tokens, and tokens inside `\boxed{}` spans and the last two layers are excluded.

F DAPO Implementation

This appendix complements Section 5 with the DAPO loss modifications relative to GRPO, the DAPO + RSP implementation, and full per-step results.

F.1 From GRPO to DAPO

The vanilla GRPO objective [Shao et al., 2024] for a group of G rollouts $\{o_i\}_{i=1}^G$ with rewards $\{R_i\}_{i=1}^G$ is

$$\mathcal{J}_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \left(\min(r_{i,t} \hat{A}_{i,t}, \text{clip}(r_{i,t}, 1-\varepsilon, 1+\varepsilon) \hat{A}_{i,t}) - \beta D_{\text{KL}}(\pi_\theta \| \pi_{\text{ref}}) \right) \right], \quad (9)$$

with importance ratio $r_{i,t} = \pi_\theta(o_{i,t} | q, o_{i,<t}) / \pi_{\theta_{\text{old}}}(o_{i,t} | q, o_{i,<t})$, group-relative advantage $\hat{A}_{i,t} = (R_i - \text{mean}(\{R_i\})) / \text{std}(\{R_i\})$, clipping range ε , and KL penalty βD_{KL} against the reference π_{ref} .

We build on the SimpleRL-Zoo [Zeng et al., 2025] training pipeline and implement DAPO on top of VeRL [Sheng et al., 2024]. DAPO modifies Eq. (9) in four ways: (i) token-level loss aggregation $1 / \sum_i |o_i|$ instead of $1 / |o_i|$, (ii) asymmetric clipping with $(\varepsilon_{\text{low}}, \varepsilon_{\text{high}}) = (0.2, 0.28)$, (iii) dual clipping safeguard $\max(L_{\text{clipped}}, c\hat{A})$ with $c = 10$, and (iv) KL terms removed (`use_kl_loss=False`, `kl_loss_coef=0`, `algorithm.kl_ctrl.kl_coef=0`).

F.2 DAPO + RSP Implementation

For DAPO + RSP, we additionally inject a group-specific RSP $\mathbf{H}_g \in \mathbb{R}^{L \times d}$ ($L = 20$) into each rollout. $\mathbf{H}_g$ is generated deterministically from a per-step group seed and injected via a forward hook; it is not a trainable parameter, so the learning signal is the policy adapting to arbitrary $\mathbf{H}_g$ rather than a learned prompt. Including $\mathbf{H}_g$ in the log-probabilities at both rollout and update time keeps the update on-policy with respect to the RSP-conditioned distribution (avoiding off-policy mismatch). With $G = 8$ and $n = 8$, each group contains a single response, so each rollout effectively sees a different $\mathbf{H}_g$.

F.3 Data, Batch, and Hyperparameters

Table 10: DAPO / DAPO + RSP training setup on Qwen2.5-Math-7B.

Field	Value
Train data	MATH Level 3–5 subset (<code>simplelr_math_35</code> , ~8,500 prompts)
Prompt template	<code>qwen-boxed</code>
Train batch size	1024
PPO mini-batch size	256 (4 PPO updates per rollout)
Rollouts per prompt G	8
Max prompt / response length	2,028 / 2,048
Rollout sampling	temperature 1.0, top- p 1.0, top- k 50
$(\varepsilon_{\text{low}}, \varepsilon_{\text{high}})$	(0.2, 0.28)
Dual clip c	10.0
Loss aggregation	token-mean ($1 / \sum_i o_i $)
KL terms	disabled
Entropy coefficient	0
Optimizer	AdamW, learning rate 5×10^{-7} (constant), no warmup
Total training	~10 epochs, ≥ 80 optimization steps
Save / eval frequency	every 10 steps
RSP (DAPO + RSP only)	suffix injection, $L = 20$, freshly drawn per rollout
Hardware	4× NVIDIA B200 GPUs
Wall-clock training time	~12 hours per run

F.4 Evaluation Protocol

Each saved checkpoint is converted to HuggingFace format and evaluated using the SimpleRL-Zoo `eval_math_nodes.sh` pipeline, which selects sampling parameters per benchmark to reduce variance on the smaller competition sets:

Table 11: Per-benchmark evaluation protocol used by SimpleRL-Zoo `sh/eval.sh`.

Benchmark	# Problems	Temperature	n_{sampling}	Metric
AIME 2024	30	1.0	32	Avg@32
AMC 2023	40	1.0	32	Avg@32
MATH-500	500	0.0	1	accuracy (mean@1)
GSM8K	1,319	0.0	1	accuracy
OlympiadBench	675	0.0	1	accuracy
Minerva Math	272	0.0	1	accuracy
GaoKao 2023-EN	385	0.0	1	accuracy

The maximum number of generated tokens is 16,000 across all benchmarks. The reported average is the unweighted mean over the seven benchmarks.

F.5 Per-Step Average Accuracy

Table 12: Four-benchmark average accuracy (%) at every checkpoint, computed over MATH-500, AIME 2024, GSM8K, and Minerva Math (the benchmarks where the two methods differ at peak). **Bold** marks each method’s peak.

Step	DAPO	DAPO + RSP
0	33.00	33.00
10	48.92	48.53
20	51.45	52.05
30	52.10	53.52
40	53.27	55.03
50	53.55	55.42
60	54.50	55.88
70	55.40	56.92
80	56.15	57.27
90	55.25	57.40
100	52.82	56.78

F.6 Limitations and Future Work

We do not claim DAPO + RSP as a complete recipe. The present comparison uses a reduced DAPO configuration that omits Dynamic Sampling and overlong reward shaping, and the evaluation runs are based on a single training seed. Confirming the gains under the full DAPO configuration, with multi-seed evaluation on small competition sets (AIME 2024, AMC 2023) where variance is highest, is left for future work.

825 G Broader Impacts

826 This work provides empirical and theoretical analysis of how random embedding injection affects
827 the reasoning behavior of large language models. The contributions are primarily foundational,
828 with potential downstream implications for reinforcement learning-based post-training of reasoning
829 models such as GRPO.

830 **Positive impacts.** RSP-induced trajectory diversity could improve the sample efficiency of RL-
831 based LLM post-training by providing richer learning signals within group-relative advantage estima-
832 tion. This has the potential to reduce compute costs associated with alignment and reasoning-oriented
833 fine-tuning, making such methods more accessible to research groups with limited resources.

834 **Potential concerns.** Methods that expand the effective output support of LLMs may increase the
835 reachability of low-probability tokens, which includes both desirable alternative reasoning paths
836 and potentially unsafe or off-distribution outputs. Practitioners applying RSP in production systems
837 should combine it with existing safety filtering and alignment mechanisms rather than treating it as a
838 standalone diversity source.

839 **Limitations on impact scope.** Since this work focuses on analysis rather than deploying a new
840 model or dataset, the direct societal impact is limited. The broader implications described above are
841 contingent on future work integrating RSP into applied training pipelines.

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